

Lecture 04

Solving Linear System I: “Good Case” of Square System

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Recall

In the last lecture ...

- Definitions of linear equations and systems of linear equations
- Definition of Matrix-vector product

Today's Lecture

Today ... More on System of **Linear Equations!**

After this lecture, you should be able to

- Tell the definition of coefficient matrix, augmented matrix
- Tell the definition of lower and upper triangular matrices
- Describe three elementary row operations and apply them to matrices
- Solve a square linear system using Gauss-Jordan Elimination [good case]

Part I

Augmented Matrix

Partly from Sec. 2.2

Length: 15 mins.

Example of Coefficient Matrix

Consider $2x_1 + 3x_2 = 3$
 $x_1 - x_2 = 4.$

coefficient matrix: $\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$

augmented matrix: $\left[\begin{array}{cc|c} 2 & 3 & 3 \\ 1 & -1 & 4 \end{array} \right]$

See general definition next.

Matrix and Linear Systems

Definition (Coefficient Matrix)

Given a linear system,

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

...

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

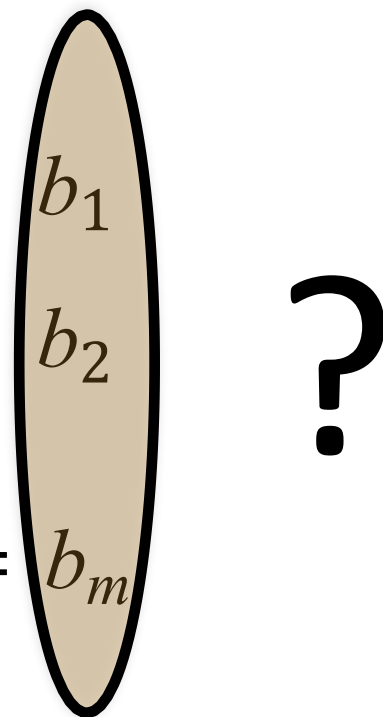
The **coefficient matrix** of the system is an $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} =: (a_{ij})_{m \times n}$$

Matrix and Linear Systems

Definition (Coefficient Matrix)

Given a linear system,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$


The **coefficient matrix** of the system is an $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} =: (a_{ij})_{m \times n}$$

Augmented Matrix

Definition (Augmented matrix)

Given a linear system,

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m\end{aligned}$$

the corresponding augmented matrix is

$$[A \mid \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]$$

Exercise

Consider

$$\begin{cases} 2x_1 - 3x_2 + 5x_3 = 7 \\ -2x_1 + x_2 + 4x_2 = 3x_3 - 1 \end{cases}$$

What is the coefficient matrix?

What is the augmented matrix?

$$\begin{bmatrix} 2 & -3 & 5 \\ 1 & 4 & -3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 5 & 7 \\ 1 & 4 & -3 & -1 \end{array} \right]$$

Overdetermined, Underdetermined and Square

Definition (System of Linear Equations)

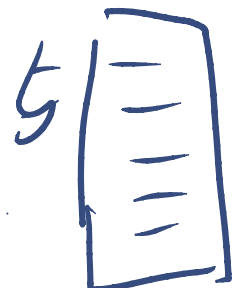
An $m \times n$ system of linear equations is

- (1) overdetermined system if $m > n$
- (2) underdetermined system if $m < n$
- (3) square system if $m = n$

Today we solve an $n \times n$ System (square)

An $m \times n$ matrix is

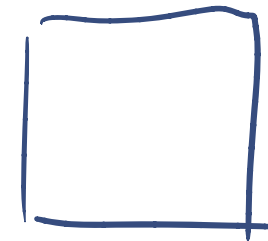
(1) tall, if $m > n$



(2) wide, if $m < n$



(3) square, if $m = n$,



Example of Special Matrices

What are special about these matrices?

$$\begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 0 & 2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} \quad \begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

Example of Special Matrices

What are special about these matrices?

$$\begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 0 & 2 & 5 \end{bmatrix}$$

All nonzero entries are below the diagonal.

All entries above the diagonal are zero.

Lower triangular matrix..

$$\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

All entries below the diagonal are zero.

Upper triangular matrix..

$$\begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} \quad \begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

All entries not on the diagonal are zero.

Diagonal matrix..

Special Matrices

Definition (Lower Triangular Matrix)

A **square** matrix of the form

$$L = \begin{bmatrix} \ell_{1,1} & & & & \\ \ell_{2,1} & \ell_{2,2} & & & \\ \ell_{3,1} & \ell_{3,2} & \ddots & & \\ \vdots & \vdots & \ddots & \ddots & \\ \ell_{n,1} & \ell_{n,2} & \dots & \ell_{n,n-1} & \ell_{n,n} \end{bmatrix}$$

$L_{in} = 0$
 $i \geq 1$
 $j = n$
 $i < j$

is called a **lower triangular matrix**.

Mathematical definition: $L_{ij} = 0$, for any $1 \leq i < j \leq n$.

Special Matrices

Definition (Upper Triangular Matrix)

A **square** matrix of the form

$$U = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} & \dots & u_{1,n} \\ & u_{2,2} & u_{2,3} & \dots & u_{2,n} \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & u_{n-1,n} \\ 0 & & & & u_{n,n} \end{bmatrix}$$

is called an **upper triangular matrix**

Mathematical definition: $U_{ij} = 0$, for any $1 \leq j < i \leq n$.

Special Matrices

Definition (Diagonal Entry) 28/10/2

For a **square** matrix $A = (a_{ij})_{n \times n}$,
each entry $a_{i,i}$ is called a diagonal entry of A .

Definition (Diagonal Matrix)

A **square** matrix $D = (D_{ij})_{n \times n}$ satisfying $D_{ij} = 0, \quad \forall i \neq j$
is called a diagonal **matrix**.

Exercise

Are these matrices:
upper triangular matrices (U), lower triangular matrices (L), diagonal matrices (D)?

$$\begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$$

L

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

D; U, L.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

D; U, L.

Part II Elementary Row Operations

Length: 30 mins.

Gaussian Elimination for 2*2 System: Matrix View

Equation view

$$\begin{cases} x_1 + x_2 = 12 & \textcircled{1} \\ 2x_1 + 4x_2 = 38 & \textcircled{2} \end{cases}$$

Solve the system below:

(Write intermediate steps)

Eliminate x_1 from $\textcircled{2}$ by $\textcircled{1}$.

$\textcircled{2} - \textcircled{1} \times 2$, get:

$$2x_2 = 14 \quad \text{---} \textcircled{3}$$

$$\Rightarrow x_2 = 7 \quad \text{---} \textcircled{4}$$

Eliminate x_2 from $\textcircled{1}$ by $\textcircled{4}$.

$\textcircled{1} - \textcircled{4}$, get

$$x_1 = 5$$

Solution.

$$\begin{cases} x_1 = 5 \\ x_2 = 7 \end{cases}$$

Augmented Matrix view

Want to represent the process in matrices.

How?

See next pages.

Operation on Rows?

Last page: Operation on equations; **translate** equations to matrices.

Can we **translate operations**
(from equation-operations to matrix-operations)?

Equation view

$$\begin{cases} x_1 + x_2 = 12 & \textcircled{1} \\ 2x_1 + 4x_2 = 38 & \textcircled{2} \end{cases}$$

Solve the system

Step 1:

$$\begin{aligned} & \textcircled{2} \quad 2x_1 + 4x_2 = 38 \\ -2 \times \textcircled{1} & \quad -2x_1 - 2x_2 = -24 \\ \hline & 0x_1 + 2x_2 = 14 \end{aligned}$$

Implicitly, we get a new system:

(often not written explicitly in the solution process)

eliminating x_1 from Eq. $\textcircled{2}$ by $\textcircled{1}$
 $\textcircled{2} - 2 \times \textcircled{1}$

Throw away x_i 's

$$\begin{array}{r} \textcircled{2} \quad [2 \quad 4 \mid 38] \\ -2 \times [1 \quad 1 \mid 12] \\ \hline [0 \quad 2 \mid 14] \end{array}$$

Augmented Matrix view

$$\left[\begin{array}{cc|c} 1 & 1 & 12 \\ 2 & 4 & 38 \end{array} \right]$$

$\Leftrightarrow$ (Row 2) - 2 * (Row 1)
in the augmented matrix

Operation on Rows: Simple Notation

What row operation?

Last page:

$$\left[\begin{array}{cc|c} 1 & 1 & 12 \\ 2 & 4 & 38 \end{array} \right]$$

Perform operation:

$$\begin{array}{r} -2 \times \left[\begin{array}{cc|c} 1 & 1 & 12 \end{array} \right] \\ + \left[\begin{array}{cc|c} 2 & 4 & 38 \end{array} \right] \\ \hline \end{array}$$

Get: $\left[\begin{array}{cc|c} 0 & 2 & 14 \end{array} \right]$

Express in row operation

$$\left[\begin{array}{cc|c} 1 & 1 & 12 \\ 2 & 4 & 38 \end{array} \right]$$

Indicate the operation:

$$R_2 \rightarrow -2R_1 + R_2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 12 \\ 0 & 2 & 14 \end{array} \right]$$

Another Row Operation

Equation view

$$x_1 + x_2 = 12 \quad \textcircled{1}$$

$$2x_2 = 14 \quad \textcircled{3}$$

Divide $\textcircled{3}$ by 2:

$$x_2 = \frac{14}{2} = 7 \quad \textcircled{4}$$

Update the linear system:

$$x_1 + x_2 = 12 \quad \textcircled{1}$$

$$x_2 = 7 \quad \textcircled{4}$$

Augmented Matrix view

$$\left[\begin{array}{cc|c} 1 & 1 & 12 \\ 0 & 2 & 14 \end{array} \right]$$

$$R_2 \rightarrow \frac{1}{2}R_2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 12 \\ 0 & 1 & 7 \end{array} \right]$$

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Matrix view.

$$\begin{cases} x_1 + x_2 = 12 & \textcircled{1} \\ x_2 = 7 & \textcircled{4} \end{cases}$$

$\textcircled{1} - \textcircled{4}$, get

$$x_1 = 5.$$

$$\begin{cases} x_1 = 5 \\ x_2 = 7 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & | & 12 \\ 0 & 1 & | & 7 \end{bmatrix}$$

$R_1 \rightarrow R_1 - R_2$ ↓

~~$R_1 \rightarrow R_1 - R_2$~~

$$\begin{bmatrix} 1 & 0 & | & 5 \\ 0 & 1 & | & 7 \end{bmatrix}$$

$$\begin{cases} x_1 + 0 \cdot x_2 = 5 \\ 0 \cdot x_1 + x_2 = 7 \end{cases}$$

Review

Solving a 2×2 system:

$$\begin{aligned} \left[\begin{array}{cc|c} 1 & 1 & 12 \\ 2 & 4 & 38 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 12 \\ 0 & 2 & 14 \end{array} \right] \\ &\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[\begin{array}{cc|c} 1 & 1 & 12 \\ 0 & 1 & 7 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & 7 \end{array} \right] \end{aligned}$$

What are the key steps?

- (1) Multiply a row by a **non-zero** scalar
- (2) Add to one row a scalar multiple of another

Third Row Operation

Equation view

$$2x_1 + 4x_2 = 38 \quad \textcircled{1}$$

$$x_1 + x_2 = 12 \quad \textcircled{2}$$

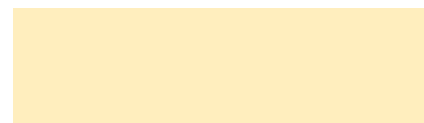
Swap the two
equations first.

$$x_1 + x_2 = 12 \quad \textcircled{2}$$

$$2x_1 + 4x_2 = 38 \quad \textcircled{1}$$

Augmented Matrix view

$$\begin{array}{c} \times (-0.5) \\ \left(\begin{array}{cc|c} 2 & 4 & 38 \\ 1 & 1 & 12 \end{array} \right) \end{array}$$



$$\begin{array}{c} \times (-2) \\ \left(\begin{array}{cc|c} 1 & 1 & 12 \\ 2 & 4 & 38 \end{array} \right) \end{array}$$

Remark:

You could eliminate Eq. ① from Eq. ②, without swapping.
In this example:

You can swap them, but not absolute necessary.

Summary of Allowable Operations on **Equations**

(1) [**Multiplication**] Multiply an equation by a non-zero scalar

$$2x_1 + 4x_2 = 38 \quad \rightarrow \quad (2\alpha)x_1 + (4\alpha)x_2 = 38\alpha, \quad \alpha \neq 0.$$

(2) [**Addition**] Add to one equation a scalar multiple of another

$$\begin{cases} x_1 + x_2 = 12, \\ 2x_1 + 4x_2 = 38 \end{cases} \quad \rightarrow \quad \begin{cases} x_1 + x_2 = 12 \\ (2+\alpha)x_1 + (4+\alpha)x_2 = 38 + 12\alpha \end{cases}$$

(3) [**Interchange**] Swap two equations

$$\begin{cases} x_1 + x_2 = 12, \\ 2x_1 + 4x_2 = 38 \end{cases} \quad \rightarrow \quad \begin{cases} 2x_1 + 4x_2 = 38 \\ x_1 + x_2 = 12 \end{cases}$$

Operations on linear equations!

Elementary **Row** Operations

Definition (Elementary Row Operations) (初等行变换)

1. **[Multiplication]** Multiply a row by a **nonzero** scalar:

$$R_i \rightarrow \alpha R_i, \quad \alpha \neq 0$$

2. **[Adding a Scaled Row]** Add a scaled multiple of one row to another row:

$$R_i \rightarrow R_i + \beta R_j$$

3. **[Interchange]** Swap the positions of two rows:

$$R_i \leftrightarrow R_j$$

Claim: Performing elementary operations will create a new system.
The new system and the original system have the same solution(s).

2nd Elementary Row Operations Preserves Solution

Claim (2nd row operation preserve solutions)

System (S1):

$$(S1) \quad \begin{cases} A_1x = b_1 & \textcircled{1} \\ A_2x = b_2 & \textcircled{2} \end{cases}$$

where $A_1, A_2 \in \mathbb{R}^{1 \times 2}$, $x \in \mathbb{R}^2$, and $b_1, b_2 \in \mathbb{R}$.

System (S2):

After performing a row operation, the system transforms into:

$$(S2) \quad \begin{cases} A_1x = b_1 & \textcircled{1} \\ (\alpha A_1 + A_2)x = \alpha b_1 + b_2 & \textcircled{3} \end{cases}$$

(S1) and (S2) have the same solutions.

What Operation Works?

Judgement:

True or False: (Multiply the coefficients of two equations)

Suppose the system of equations is given by:

$$(P1) \quad \alpha_1 x_1 + \alpha_2 x_2 = \underline{b_1},$$

$$(P2) \quad \beta_1 x_1 + \beta_2 x_2 = \underline{b_2}.$$

After performing the row operation:

$$(P1') \quad \alpha_1 x_1 + \alpha_2 x_2 = b_1,$$

$$(P2') \quad (\alpha_1 \beta_1) x_1 + (\alpha_2 \beta_2) x_2 = \alpha_1 b_2,$$

we claim that $(P1)$ and $(P2)$ have the same solutions as $(P1')$ and $(P2')$.

Find a counter-example.

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$$(x_1^*, x_2^*) = (1, -1)$$

⇓

$$\begin{array}{l} (p_1) \\ (p_2) \end{array} \left\{ \begin{array}{l} x_1 + x_2 = 0 \\ 2x_1 + x_2 = 1 \end{array} \right.$$

$$(p_1') \quad x_1 + x_2 = 0$$

$$(p_2') \quad 2x_1 + x_2 = 0$$

$$\text{solution} \quad \left\{ \begin{array}{l} x_1 = 1 \\ x_2 = -1 \end{array} \right.$$



so the statement is false.

Other Operations

Exercise (Other Operations)

Can the following operation be performed when solving a linear system??

Multiply a row by zero

No

Part III High-level Summary of Gauss- Jordan Elimination

Length: 20 mins.

Gauss-Jordan Elimination: 3 by 3 System. Phase I

Phase 1: Forward Elimination (Equation)

$$\begin{cases} x + y + z = 6 & (1) \\ x + 2y + 2z = 9 & (2) \\ x + 2y + 3z = 10 & (3) \end{cases}$$

$(2) - (1), (3) - (1)$, get

$$y + z = 3 \quad (4)$$

$$y + 2z = 4 \quad (5)$$

$(5) - (4)$, get

$$z = 1 \quad (6)$$

$$\begin{cases} x + y + z = 6 & (1) \\ y + z = 3 & (4) \\ z = 1 & (6) \end{cases}$$

Phase 1: Forward Elimination (Matrix)

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 2 & 9 \\ 1 & 2 & 3 & 10 \end{array} \right]$$

~~$R_2 = R_2 - R_1$~~

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & 2 & 4 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

Gauss-Jordan Elimination: 3 by 3 System. Phase I

Phase 2: Backward Substitution (Equation)

$$x + y + z = 6 \quad (1)$$

$$y + z = 3 \quad (4)$$

$$z = 1 \quad (6)$$

(1) - (6), (4) - (6), get

$$\begin{cases} x + y = 5 & (7) \\ y = 2 & (8) \end{cases}$$

(7) - (8): $x = 3 \quad (9)$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \quad \begin{matrix} (9) \\ (8) \\ (6) \end{matrix}$$

← one more step

Phase 2: Backward Substitution (Matrix)

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 - R_3$$

$$R_2 \rightarrow R_2 - R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

GJ-E for General Matrix: First Step

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array} \right] \longrightarrow \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ 0 & \hat{a}_{22} & \cdots & \hat{a}_{2n} & \hat{b}_2 \\ 0 & \hat{a}_{32} & \cdots & \hat{a}_{3n} & \hat{b}_3 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \hat{a}_{n2} & \cdots & \hat{a}_{nn} & \hat{b}_n \end{array} \right]$$

Identify the leading entry in the first row, a_{11} and ensure it is non-zero.

Assumption for now (good case): a_{11} is nonzero.

Definition (informal) (Pivot)

First non-zero in the row that does the elimination

Use a_{11} as the pivot to eliminate all entries below it in the first column.

Perform row operations:

$$R_i \rightarrow R_i - \frac{a_{i1}}{a_{11}} R_1 \quad \text{for } i = 2, 3, \dots, m.$$

GJ-E for General Matrix: Second Step

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ 0 & \hat{a}_{22} & \cdots & \hat{a}_{2n} & \hat{b}_2 \\ 0 & \hat{a}_{32} & \cdots & \hat{a}_{3n} & \hat{b}_3 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \hat{a}_{n2} & \cdots & \hat{a}_{nn} & \hat{b}_n \end{array} \right] \longrightarrow \left[\begin{array}{cccc|c} a_{11} & \times & \times & \cdots & \times \\ 0 & \hat{a}_{22} & \times & \cdots & \hat{b}_2 \\ \vdots & 0 & \times & \ddots & \vdots \\ 0 & 0 & \times & \cdots & \times \end{array} \right]$$

1. **Identify the pivot:** The pivot for this step is $\hat{a}_{22}$, the updated entry in the second row and second column.
2. **Eliminate entries below the pivot:** Use $\hat{a}_{22}$ to eliminate all entries below it in the second column:

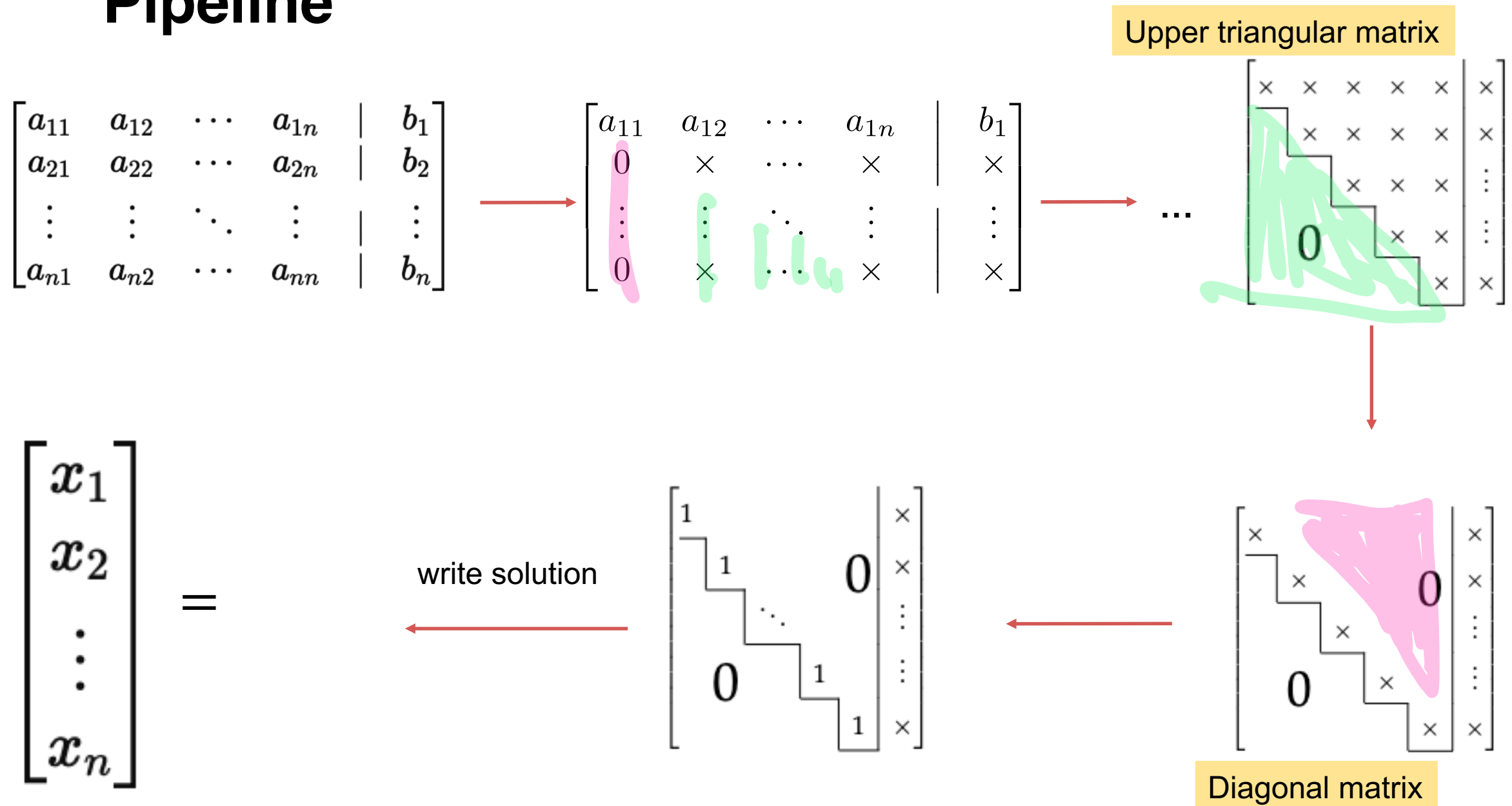
$$R_i \rightarrow R_i - \frac{\hat{a}_{i2}}{\hat{a}_{22}} R_2 \quad \text{for } i = 3, 4, \dots, m$$

3. **Update the matrix:** The rows below the second row are updated accordingly, resulting in a modified matrix.

Assumption for now (good case): $\hat{a}_{22}$ is nonzero.

Gaussian-Jordan Elimination for “Good” Systems

Pipeline



(Think: does it always work?)

Summary of GJ-E for Solving Square Systems

Phase 1: Forward Elimination.

Perform elementary row operations and try to get an **upper triangular** matrix.

Phase 2: Backward substitution

Perform elementary row operations and try to get a **diagonal matrix**.

Assumption 1

At each iteration of the forward elimination, the next diagonal entry is nonzero.

Claim 1 Under Assumption 1, we can get a diagonal matrix at the end of Step 2.

Corollary 1 Under Assumption 1, the system has a unique solution.

This assumption may not hold for some problems; will discuss later.

Concluding Section



Summary Today

One sentence summary:

Detailed summary: